

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP packet was sent from the source IP address, 192.51.100.15.52 to the DNS server (IP address 203.0.113.2) at port 53 to retrieve the IP address of yummyrecipesforme.com. This results in a ICMP error message from the indicating that the UDP port 53 is unreachable. UDP port 53 is the primary port for DNS queries.

The logs also reflect a query identification number, 35084 and a plus sign after it. This indicates there are flags associated with the UDP message. The "A?" indicates a flag associated with the DNS request for an A record which maps a domain name to an IP address.

The same request is attempted two more times, both resulting in the same ICMP error message. Therefore, the most likely issue is that the DNS server is down.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time incident occurred: 1:24 pm and 32s.

Several customers reported that they were not able to access the client company website www.yummyrecipesforme.com, and saw the error "destination port unreachable" after waiting for the page to load.

The team tested the same URL and used a network packet analyzer, tcpdump to check and sent a request. The response was an ICMP packet that noted that port 53 was unreachable. This was tested two more times, resulting in the same outcome.

The next step is to identify whether the outage of the DNS Server was caused by a successful attack or a misconfiguration on the firewall, such as traffic to port 53 being blocked. Security engineers are already looking into the root cause of the situation.